

But what about a worse way... *

* (... of packing squares into a larger shape [†])

[†](I actually was tempted at fitting as many P's in the title as possible, with the working title being "Provable Penalties Produce Pessimistic Packings" but was advised against this. Although I was also advised against having a sub-sub title explaining this.)

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Abstract—While decades of computational geometry have focused on minimizing spatial footprints, this paper bravely asks: "What if we just made it worse?" We formalize the novel field of Pessimal Packing, establishing a rigorous mathematical taxonomy of adversarial spatial inefficiencies for hinged polyominoes. By evaluating continuous kinematic configurations of $N = 3$ unit squares against varying container constraints, we catalog the absolute geometric limits of spatial waste. We mathematically prove that a naive, axis-aligned bounding square permits a symmetric $\arctan(0.5)$ staircase yielding an area of 10.472. By deploying a Differential Evolution heuristic to solve the non-differentiable domains of rotationally-hostile evaluators, we establish the global maxima for the remaining spatial paradigms: the arbitrary bounding rectangle ($A \approx 7.089$), the invariant bounding square ($A \approx 6.877$), and the container-free maximum convex hull ($A \approx 5.202$). Together, these results map the exact, often irrational geometries required to be as pathologically uncooperative as mathematically possible.

I. INTRODUCTION

Ever since Erdős and Graham sought to minimize wasted space in square packing [2], the scientific community has been plagued by an obsession with efficiency. While Friedman has extensively documented the optimal bounding containers for unit squares [1] —achieving widespread validation from the world's most rigorous peer-review body, the internet, the literature is shockingly silent on how to pack them as terribly as possible. To correct this historical oversight and mathematically guarantee maximum spatial waste, we deployed the Differential Evolution heuristic [3], heavily penalizing the algorithm whenever it accidentally discovered a useful packing arrangement.

II. DEFINING THE OBJECTIVE

As we yearn for maximized inefficiencies, we must first come up with a rigorous definition of our problem. We start by simply inverting the original packing puzzle, without adding any kinematic constraints:

Problem 1: Find the arrangement of N unit squares for which the area of the smallest surrounding square is maximized.

This, however, introduces several existential flaws, which we can formalize into:

Theorem 1: Problem 1 isn't very interesting. ¹

Proof: A mathematical problem is widely considered interesting only if it is *well-posed* in the sense of Hadamard [4], which strictly requires that a valid solution actually exists. Because Problem 1 imposes no topological connectivity constraints on the N unit squares, one can simply place them infinitely far apart. The bounding area grows monotonically with the distance between the disjoint squares, approaching infinity ($A \rightarrow \infty$). Consequently, no maximal arrangement exists, rendering the problem both mathematically ill-posed and comedically unfulfilling.² ■

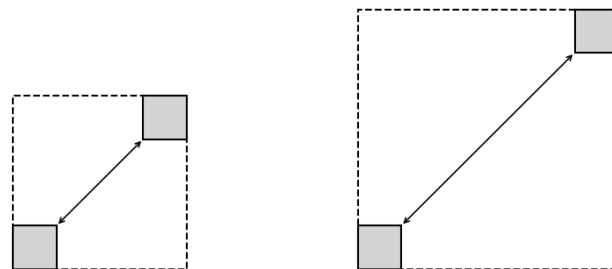


Fig. 1. Two attempted solutions for Problem 1, one slightly less suboptimal than the other.

We follow that any *well-posed* variation of Pessimal Packing must introduce topological constraints:

Problem 2: Find the arrangement of N connected unit squares for which the area of the smallest axis-aligned surrounding square is maximized.

III. SOLVING THE BOUNDING BOX SPACE³

¹Citation needed

²Never-mind, shouldn't have asked.

³This entire section isn't very interesting.[†]

[†] citation needed [‡]

[‡] it's the math part, there is never anything interesting in the math part.

To isolate the theoretical maximum for $N = 3$, we must distribute the finite "stretch budget" of the polyomino across both the X and Y axes. By anchoring a central square and mirroring the rotation of the two peripheral squares across the primary diagonal, we ensure the resulting bounding box expands symmetrically.

As illustrated in Fig. 2, this creates a clear kinematic progression. A straight linear chain (Configuration A) heavily biases one axis, while an aggressive 45° split (Configuration B) over-corrects, sacrificing too much distal reach. The true spatial maximum must lie somewhere between these two intuitive extremes.

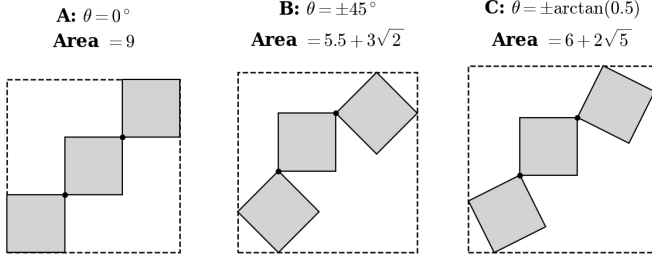


Fig. 2. Comparison of candidate solutions for Problem 2, with Configuration C representing the optimal peripheral sprawl.

With this visual intuition established, we now formalize the exact mathematical peak of this symmetric diagonal projection.

Theorem 2: For $N = 3$ corner-connected ⁴ unit squares, a symmetric diagonal projection maximizes its bounding box area strictly at the peripheral rotation angle of $\theta = \arctan(0.5)$.

Proof: Let an unrotated peripheral square, Q_{tr} , occupy the space $x \in [1, 2]$ and $y \in [1, 2]$. It is hinged at $(1, 1)$ and rotated clockwise by an angle θ . We restrict the domain to $\theta \in [0, \pi/2]$ to represent outward peripheral sprawl; angles outside this domain result in physical self-intersection or trivial area reduction. The extreme coordinate bounds of Q_{tr} relative to its hinge point extend into the positive space by $\Delta x_{tr} = \cos \theta + \sin \theta$ and $\Delta y_{tr} = \cos \theta$.

To enforce diagonal symmetry, a second peripheral square, Q_{bl} (initially occupying $x \in [-1, 0]$ and $y \in [-1, 0]$), is hinged at $(0, 0)$ and mirrored by rotating it counter-clockwise by θ . Its bounds extend into the negative space by $\Delta x_{bl} = -\cos \theta$ and $\Delta y_{bl} = -(\cos \theta + \sin \theta)$.

The total width $W(\theta)$ of the bounding box is defined by the difference between the maximal and minimal x -coordinates:

$$W(\theta) = (1 + \cos \theta + \sin \theta) - (-\cos \theta) = 1 + 2\cos \theta + \sin \theta$$

Because the configuration is mirrored across the primary diagonal ($y = x$), the height $H(\theta)$ is identical to $W(\theta)$,

⁴We trivially dismiss edge-to-edge connections. Connecting squares along their edges truncates the internal diagonal displacement ($\sqrt{2}$), strictly reducing the kinematic reach and bounding the polyomino by a smaller topological diameter. Such arrangements are categorically excluded from serious inquiry.

making the area $A(\theta) = W(\theta)^2$. To maximize this strictly positive function, we apply the harmonic addition identity ($a \cos \theta + b \sin \theta = R \cos(\theta - \alpha)$), where $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = \frac{b}{a}$.

Substituting $a = 2$ and $b = 1$, the width function algebraically collapses to:

$$W(\theta) = 1 + \sqrt{5} \cos(\theta - \arctan(0.5))$$

Because the global maximum of any unscaled cosine function is exactly 1 (occurring when its internal argument is 0), the maximum width is strictly achieved when $\theta - \arctan(0.5) = 0$. Thus, the optimal geometry is uniquely locked at $\theta = \arctan(0.5)$. ■

While this establishes the bounds for $N = 3$, we must formally address the behavior of this projection as the total number of squares N approaches infinity.

Theorem 3: The optimal peripheral sprawl angle of $\theta = \arctan(0.5)$ is strictly invariant to the arbitrary elongation of the central diagonal chain.

Proof: Let us construct a geometrically generalized configuration by inserting a chain of k additional, unrotated unit squares between the two peripheral bounding squares Q_{tr} and Q_{bl} , such that the total number of squares is $N = k + 3$.

Because these internal squares are axis-aligned and rigidly connected corner-to-corner along the $y = x$ vector, they form a simple, monotonic staircase. The central chain now consists of $1 + k$ squares, each contributing exactly 1 unit of distance to both the total width and the total height of the bounding box.

The generalized side length function becomes:

$$W_k(\theta) = (1 + k) + 2\cos \theta + \sin \theta$$

To find the optimal rotation for the peripheral squares in this elongated chain, we differentiate with respect to θ . Because the linear contribution of the central chain ($1 + k$) is a geometric constant wholly independent of the peripheral rotation θ , it vanishes entirely upon differentiation:

$$\frac{d}{d\theta} W_k(\theta) = 0 - 2\sin \theta + \cos \theta = 0$$

The critical point remains isolated identically at $\tan \theta = 0.5$. Thus, the optimal angle to maximize the bounding area of a diagonally-symmetric polyomino chain is a universal geometric constant, regardless of the internal distance separating the peripheral hinges. ■

IV. FINDING BETTER PROBLEMS

Having established the monotonic elegance of the axis-aligned bounding square, we encounter a philosophical crisis: the resulting maximization is, frankly, too orderly. In the pursuit of true spatial waste, an elegant solution is a failed solution.

We hypothesize an inverse relationship in computational geometry: if an inherently interesting minimization problem

yields a predictable maximization, then a problem for which the minimization is entirely trivial might harbor a pathologically complex maximization. To test this, we relax the container constraint from a strict square to an arbitrarily rotated bounding rectangle.

Problem 3: Find the arrangement of N connected unit squares for which the minimum possible area of an enclosing rectangle is maximized.

To understand the adversarial nature of this formulation, we must first observe its minimization. As demonstrated in Fig. 3, minimizing the bounding rectangle of $N = 3$ squares trivially collapses into a contiguous 3×1 linear chain.

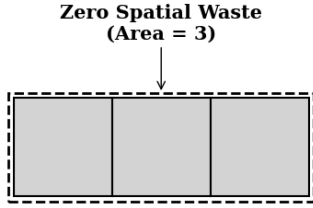


Fig. 3. The trivial global minimum for rectangular packing: a strictly linear 3×1 alignment yielding an optimal area of exactly 3.

Because the minimal area strictly matches the combined area of the polyominoes themselves ($A = 3$), there is absolutely zero spatial waste. It is the epitome of packing efficiency, and therefore, our ultimate enemy.

We also formulate a narrower version of this problem:

Problem 4: Find the arrangement of N connected unit squares for which the minimum possible area of an enclosing rectangle is maximized, and for which this rectangle is a square.

Or generalize to other shapes ⁵

Problem 5: Find the arrangement of N connected unit squares for which the area of its convex hull is maximized.

V. SOLVING BETTER PROBLEMS

No longer constrained by the rigid $1 : 1$ aspect ratio of the bounding square, Problem 3 presents a more complex landscape of solutions.

If we subject our previously optimal $\arctan(0.5)$ staircase to this metric, the container simply collapses around the polyomino's primary axis of elongation. Because the shape is inherently narrow along its diagonal, its true minimum enclosing area catastrophically drops. To survive this objective metric, the polyomino can no longer rely on a single, elongated sprawl; it must construct a configuration that is pathologically resistant to rotation, remaining uniformly wide from every conceivable angle.

This transforms the objective into a hostile minimax formulation: the polyomino's hinges must be configured to maximize

⁵For circles this problem is again trivial, while all other shapes not mentioned, including but not limited to parallelograms, triangles, pentagons, hexagons and ellipses are left for future work.

a bounding area that the container metric actively seeks to minimize.

A. Adversarial Optimization via Differential Evolution

Because the minimum enclosing rectangle is a non-differentiable function of the hinge angles, closed-form calculus cannot locate the global maximum. To navigate this rugged fitness landscape, we deployed a Differential Evolution (DE) algorithm [3].

The genome of the optimizer consisted of the exact continuous perimeter placements and rotational offsets of the peripheral squares.

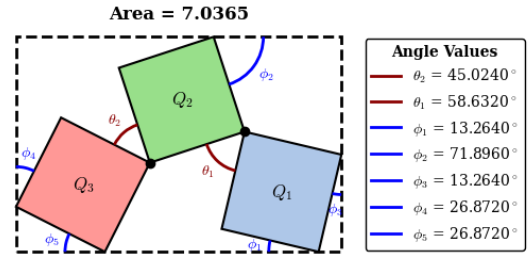


Fig. 4. Local Problem 3 optimum and convergence trap with $A \approx 7.0365$.⁷

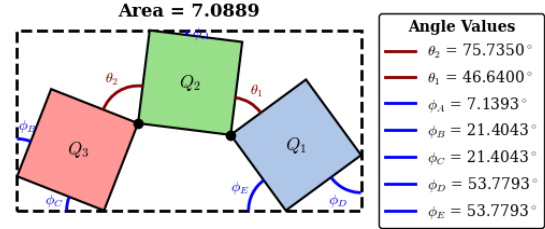


Fig. 5. Global Problem 3 maximum with $A \approx 7.0889$.

Evaluation over 10,000 runs demonstrates that the optimizer does not always converge on the global maximum. The ruggedness of the terrain causes approximately 80% of all runs to get stuck in local optima resembling the trap shown in Fig. 4, with only the remainder breaking through to the optimal configuration shown in Fig. 5.

Turning to Problem 4, we enforce the rigid invariant square constraint by introducing a heavy penalty if the aspect ratio of the bounding box deviates from $1 : 1$. This yields the configuration in Fig. 6.

Here, optimality also appears to have been definitively achieved. The polyomino shifts its hinges to ensure that the area of its smallest surrounding square is perfectly equal to its absolute minimum bounding rectangle, eliminating any rotational "shortcuts" for the container (Fig. 7).

⁷And you thought this paper was in black and white!

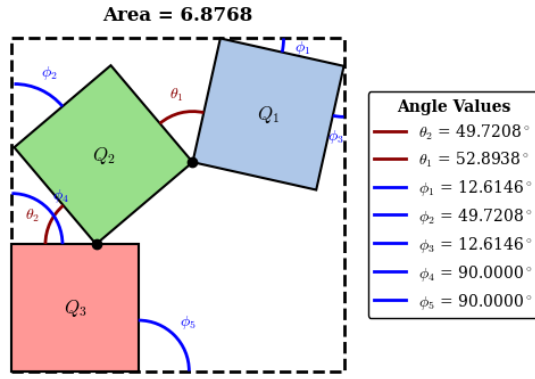


Fig. 6. Global Problem 4 maximum with $A \approx 6.8768$.

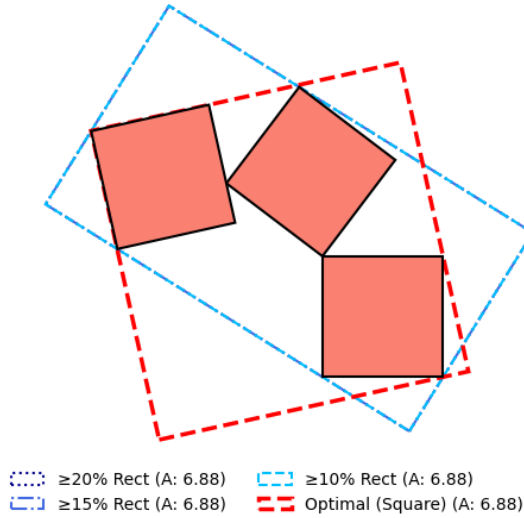


Fig. 7. Showcase of square and non-square bounding rectangles for the Problem 4 solution, confirming the 1 : 1 aspect ratio convergence.

B. The Convex Hull Concession

Finally, to appease the inevitable demands of peer review⁸, we also include a solution to Problem 5, which strips away the container entirely in favor of the maximum convex hull.

Deprived of an orthogonal container to actively fight against, the shape loses its irrational, asymmetrical defenses. It unfortunately collapses back into perfect, elegant symmetry (Fig. 8), forming a crescent that traps empty space by connecting at a mirrored 47.12° angle at both junctions.

VI. FURTHER WORK

Future research in this domain requires analyzing solutions beyond $N = 3$ for each of the proposed problems. A primary objective will be discovering the critical value of N at which, if ever, the polyominoes transition from simple sprawling arcs to other configurations, as well as formalizing the mathematical proofs for the exact irrational angles discovered empirically here.

⁸People from the math department said this is interesting, and should be included

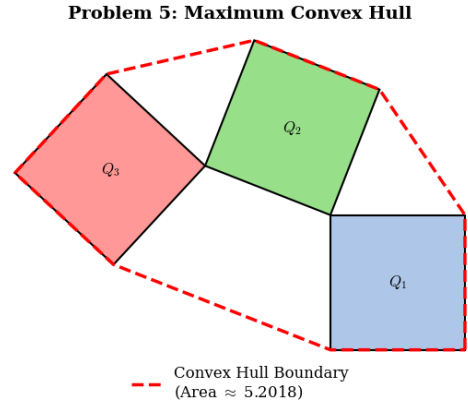


Fig. 8. Global Problem 5 maximum with $A \approx 5.2018$. The squares form a perfect crescent, touching under a 47.12° angle at both connection points.

VII. PRACTICAL APPLICATIONS

While the pursuit of pessimal packing may initially appear to be a purely theoretical exercise in mathematical masochism, it possesses profound implications for modern commercial industry, particularly in the domain of consumer packaging and retail shelf presence.

A. (Mis-) use of proven guarantees

In recent years, the consumer goods sector—spanning luxury chocolates, cosmetics, and individually wrapped snacks—has increasingly relied on “shrinkflation” to maintain profit margins. This necessitates maximizing the perceived volume of the external packaging while minimizing the actual mass of the product inside. However, manufacturers frequently face class-action litigation over “non-functional slack-fill,” defined legally as empty space within a package that serves no legitimate purpose.

The adversarial limits established in this paper provide a robust, mathematically weaponized defense against such litigation.

By deliberately molding or arranging a product into our $A \approx 7.089$ configuration (Fig. 5) or the $A \approx 6.877$ rotation-invariant square (Fig. 6), a corporation can drastically inflate the required size of its packaging. When challenged by consumer protection agencies regarding the absurd amount of empty air in the box, the corporate legal team can cite our findings to prove that the oversized box is not a deceptive marketing tactic, but rather the strictly proven *minimum possible bounding rectangle* required to house the product’s aesthetic shape.

B. Aesthetic Deterrence in Academia

Finally, the topologies discovered herein hold significant promise in the field of social deterrence. By printing the $A \approx 7.089$ configuration on a standard garment (e.g., a T-shirt), the wearer generates an immediate, visceral repulsion field against humorless mathematicians. The sheer, provable inefficiency of these packings is theoretically sufficient to terminate unwanted

networking interactions at academic conferences before they begin, acting as a highly effective visual protection charm for those who appreciate recreational mathematical anarchy.

VIII. ACKNOWLEDGMENTS

Thanks to my online friends (who wish to remain anonymous) who found this problem as irresistibly cursed as I did, and who suffered through the erroneous outputs of awful solvers dreamt up in the middle of the night.

Special thanks also goes to my Advisor and my Lab, the computational power of which was **not**—and I reiterate, was **not**—used for this project.

A final thanks goes to everyone who listened to me ramble about this, willingly or otherwise, or who suffered second-hand during the writing process. This includes, but is not limited to, my online friends, my offline friends, my family, and, once again, everyone in my lab.

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